

Interim Report: Analysing Daily Weather Data from Selected Indian Cities, 2000-2024

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Background

India has experienced tremendous development and population growth over the past 25 years and continues to do so today as a global economic powerhouse. With the second-largest population in the world in a relatively small area, facing staggering levels of socioeconomic inequality, India has become both a contributor to and a victim of the climate crisis. The country’s long, densely populated coastline and warm climate make it especially vulnerable to catastrophic natural phenomena such as extreme heat, drought and tropical storms.

Data Source

The dataset, India Daily Weather (2000-2024) – Major Cities, was retrieved from [Kaggle](#).

This dataset contains daily weather information from 10 major cities in India between January 1, 2000 and December 31, 2024. Each daily record includes minimum and maximum temperature, as well as perceived temperature; maximum wind and wind gust speeds; precipitation levels and WMO weather codes.

With this dataset, I hope to find insights on general, regional patterns in India’s weather, as well as identify trends from the past 25 years and potentially future ones.

The dataset consists of a single csv file with 12 columns and over 9000 rows.

Data Profile

india_daily_weather_2000_2024.csv

Key	Column Name	Data Type	Description
	city	object	Name of Indian city in which weather was recorded.
K	date	object	Date of weather record (YYYY-MM-DD).
	temperature_2m_max	float64	Maximum temperature, measured at 2m height (°C).
	temperature_2m_min	float64	Minimum temperature, measured at 2m height (°C).
	apparent_temperature_max	float64	Maximum perceived (feels-like) temperature, affected by wind and humidity (°C).
	apparent_temperature_min	float64	Minimum perceived (feels-like) temperature, affected by wind and humidity (°C).
	precipitation_sum	float64	Total daily precipitation (mm).
	rain_sum	float64	Total daily rainfall (mm)
	weather_code	int64	WMO weather-interpretation code.

	wind_speed_10m_max	float64	Maximum wind-speed, measured at 10m height (km/h).
	wind_gusts_10m_max	float_64	Maximum wind gust speed, measured at 10m height (km/h).
	wind_direction_10m_dominant	int64	Dominant wind direction (degrees).

Initial Data Wrangling and Consistency Checks Performed

- No nulls, duplicates or mixed-type columns found.
- Converted the date column to datetime format.
- Created a column (wmo_description) containing text descriptions of WMO weather codes, retrieved from [official website](#).
- Created columns for mean daily temperatures (temperature_2m_mean and apparent_temperature_mean)
- Created columns for month and year, as well as for month and year (e.g. 05-2010 for May 2010), for the purpose of aggregating daily data by each month.
- Created columns showing the difference between apparent and real temperature (temp_diff_min, temp_diff_max and temp_diff_mean). This was done by subtracting the actual temperature from its corresponding apparent temperature. For example, temp_diff_min for a certain city and date can be found by subtracting temperature_2m_min from apparent_temperature_min.
- Created aggregate columns showing the monthly average of the following variables: minimum, mean and maximum daily temperature; minimum, mean and maximum apparent temperature; minimum, mean and maximum temperature difference; precipitation, rain, maximum wind-speed, maximum wind-gust speed.

Column Derivations

Column Name	Original Column Derived From	Conditions
temperature_2m_mean	temperature_2m_min, temperature_2m_max	Finds the mean value between the two temperatures for each row.
apparent_temperature_mean	apparent_temperature_min, apparent_temperature_max	Finds the mean value between the two apparent temperatures for each row.
year	date	Shows the year for each row.
month	date	Shows the month for each row.
month_year	month, year	Combines month and year for each row.
min_temp_monthly_avg,	temperature_2m_min, month_year, city	Aggregates daily minimum temperature by month for each year, for each city.
max_temp_monthly_avg,	temperature_2m_max, month_year, city	Aggregates daily maximum temperature by month for each year, for each city.
mean_temp_monthly_avg	temperature_2m_mean, month_year, city	Aggregates daily mean temperature by month for each year, for each city.

apparent_min_temp_monthly_avg	apparent_temperature_min, month_year, city	Aggregates daily apparent minimum temperature by month of each year for each city.
apparent_max_temp_monthly_avg	apparent_temperature_max, month_year, city	Aggregates daily apparent maximum temperature by month of each year for each city.
apparent_mean_temp_monthly_avg	apparent_temperature_mean, month_year, city	Aggregates daily apparent mean temperature by month of each year for each city.
precipitation_monthly_avg	precipitation_sum, month_year, city	Aggregates daily precipitation levels by month of each year for each city.
rain_monthly_avg	rain_sum, month_year, city	Aggregates daily rain levels by month of each year for each city.
max_wind_monthly_avg	wind_speed_10m_max, month_year, city	Aggregates daily maximum wind speeds by month of each year for each city.
max_windgusts_monthly_avg	wind_gusts_10m_max, month_year, city	Aggregates daily maximum wind-gust speeds by month of each year for each city.
temp_diff_min	apparent_temperature_min, temperature_2m_min	Subtracts the actual daily minimum from the apparent daily minimum to find the difference in perceived temperature.
temp_diff_max	apparent_temperature_max, temperature_2m_max	Subtracts the actual daily maximum from the apparent daily maximum to find the difference in perceived temperature.
temp_diff_mean	apparent_temperature_mean, temperature_2m_mean	Subtracts the actual daily mean from the apparent daily mean to find the difference in perceived temperature.
temp_diff_min_monthly	temp_diff_min, month_year, city	Aggregates the daily minimum temperature difference by month for each year, for each city.
temp_diff_max_monthly	temp_diff_max, month_year, city	Aggregates the daily maximum temperature difference by month for each year, for each city.
temp_diff_mean_monthly	temp_diff_mean, month_year, city	Aggregates the daily mean temperature difference by month for each year, for each city.
date_numeric	date	Using matplotlib date2num, creates a unique date identifier in numeric format, suitable for certain types of visualizations.
wmo_description	weather_code	Assigns text descriptions, retrieved from the WMO official website , for each WMO weather code.

Questions to Explore

- What kinds of seasonal patterns in temperature, wind and precipitation can be observed in each city? Which cities share similar seasonal weather patterns?
- What kind of, if any, changes can be observed in the overall weather conditions and those of individual cities from 2000 to 2024?
- Which cities experienced extreme weather conditions (max/min temperature, rainfall, wind speeds) during this period? Do these cities have any other common factors that might affect or contribute to these weather conditions?
- How does the difference between apparent and real temperature vary by city, seasonally and over time? What factors affect apparent temperature and how?

Summary Statistics

- Maximum daily temperatures range from 12.3°C to 46.4°C, with a mean of 30.9°C. Minimum daily temperatures range from 1.7°C to 34.4°C with a mean of 21.1°C. Mean daily temperatures range from 7.65°C to 39.65°C with a mean of 26°C.
- The maximum amount of precipitation recorded in a single day is 339.9mm. Most days across all cities report no precipitation at all.
- Maximum wind speeds measured at 10 metres range from 3.6km/h to 68.9km/h, with a mean value of 15.5km/h. Maximum wind-gust speeds range from 10.4km/h to 142.9km/h with a mean value of 32.2km/h.
- The lowest minimum daily temperature was 1.7°C, recorded in Lucknow, on December 30, 2019.
- The highest maximum daily temperature was 46.4°C, recorded in Ahmedabad on May 23, 2024, and in Delhi on May 28, 2024.
- The lowest apparent minimum temperature was -1.1°C, recorded in Jaipur on December 31, 2019.
- The highest apparent maximum temperature was 49.4°C, recorded in Lucknow on May 30, 2024.
- The highest rainfall was 339.9mm, recorded in Ahmedabad on July 22, 2017.
- The highest measured wind speed was 68.9km/h, recorded in Kolkata on May 20, 2020.
- The highest wind gust speed was 142.9km/h, recorded in Kolkata on May 20, 2020.

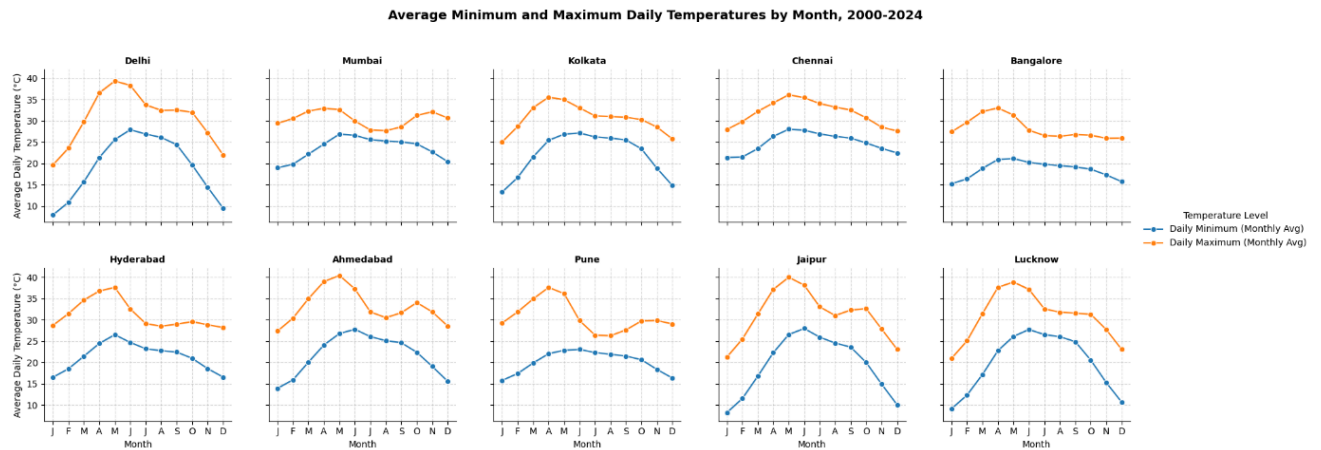
Observations based on top and bottom 5 values

- All lists of top and bottom 5 values can be found in the script data-prep-EA-visualization.
- Lucknow and Jaipur were affected by especially cold temperatures between 30 and 31 December, 2019. Jaipur and Delhi also experienced unusually cold temperatures around the same time of the year in 2011 and 2013.
- Late May 2024 was the hottest season on record, with Delhi, Ahmedabad and Jaipur being particularly affected.
- Jaipur experienced the coldest apparent temperatures out of all cities, particularly around the beginning of the year.
- While Lucknow did not record the highest temperatures in 2024, it did record the highest apparent temperatures at the same time of the year. This indicates that the northern part of the country experienced a severe heatwave at that time.
- Ahmedabad and Mumbai recorded 4 out of the 5 highest rainfall levels. Chennai's high rainfall record appears in December, unlike the other 4 records which appear during the summer and late summer months.
- Kolkata experienced abnormally high wind speeds and wind gusts on May 20, 2020. Chennai's highest wind speed and wind gust speed were recorded in December 2016. The same city recorded one of the highest rainfall levels in December 2023. These indicate that Chennai commonly experiences high levels of wind and rain during this time of year.
- The cities of Bangalore, Hyderabad and Pune do not appear in any of these lists, indicating that these cities are less prone to extreme weather events.

General Seasonal Patterns, All Cities

The following charts aggregate all values by month and show the typical seasonal weather patterns of each city over a single year.

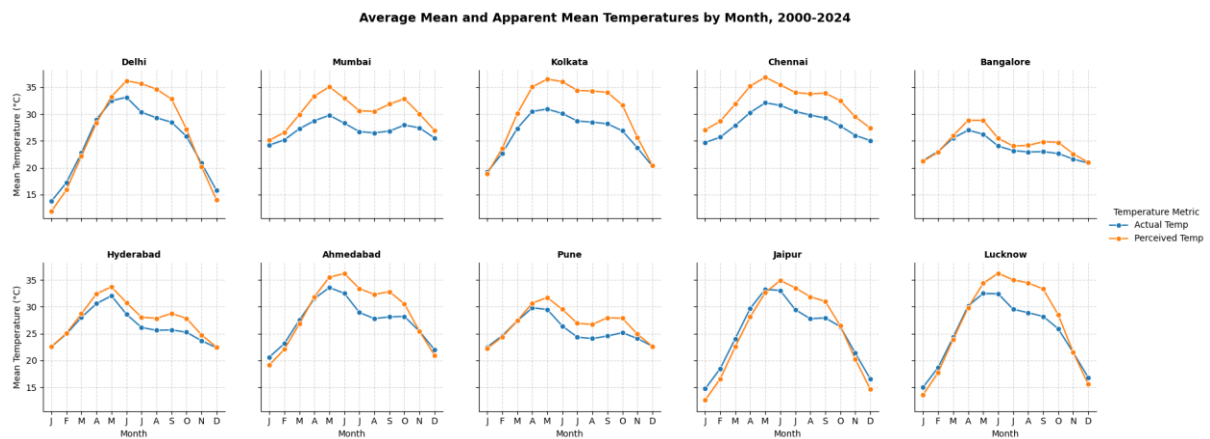
Daily Minimum and Maximum Temperatures by Month, All Cities, 2000-2024



Observations:

- All cities have daily minimum temperatures that peak during the summer months.
- With the exception of Chennai, all cities show a drop in daily maximum temperatures between May and October. This drop is especially pronounced in Mumbai, Ahmedabad, Pune and Jaipur.
- Between the months of June and September, daily minimum and maximum temperatures in Mumbai are particularly close together.

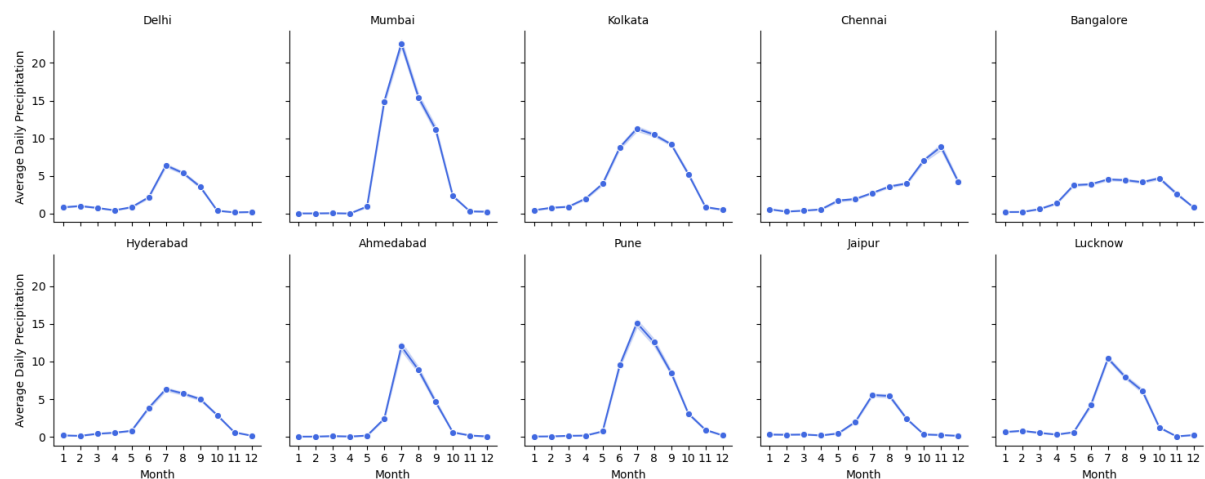
Actual vs Apparent Daily Mean Temperatures by Month, All Cities, 2000-2024



Observations

- Only the cities of Delhi, Ahmedabad, Jaipur and Lucknow show perceived temperatures that are lower than actual temperatures.
- All cities show higher perceived temperatures between the months of May and October.
- Mumbai and Chennai show perceived temperatures consistently higher than actual ones.
- Bangalore, Hyderabad and Pune show the least difference between real and perceived temperatures overall. Kolkata, Mumbai and Chennai show the greatest difference between these two temperature measures.

Precipitation Levels, All Cities, 2000-2024

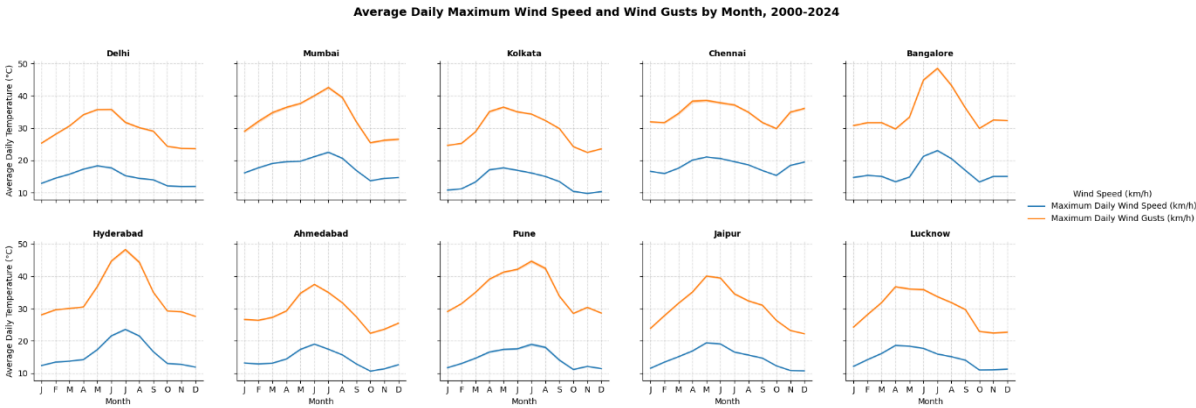


Observations:

- Mumbai typically receives the highest amounts of precipitation each year. Delhi, Bangalore, Hyderabad and Jaipur receive the lowest.
- All cities, except for Chennai, show increased precipitation between the months of May and October, peaking in July. In Chennai, precipitation increases between September

and December, peaking in November. Bangalore receives increased precipitation during both these periods, indicating the effect of different regional rainy (monsoon) seasons.

Maximum Wind and Wind Gust Speeds, All Cities, 2000-2024

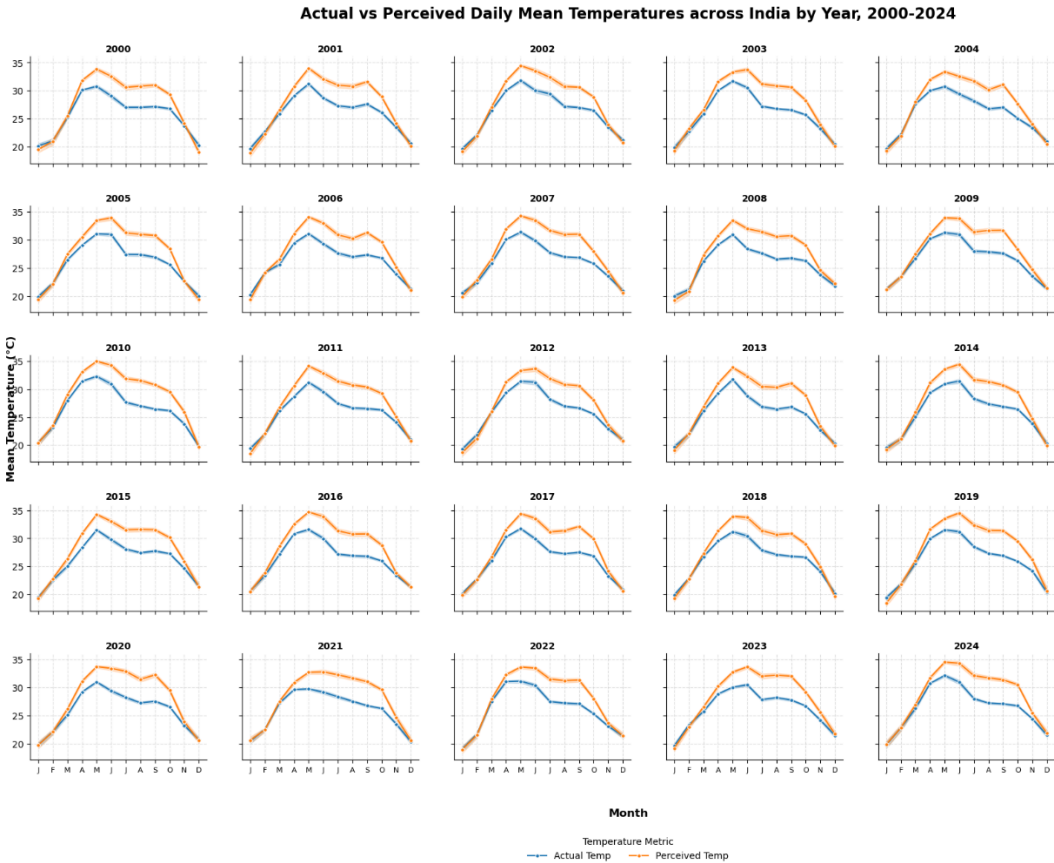


Observations

- Wind speeds increase in all cities during the summer months. This increase is especially pronounced in Bangalore and Hyderabad. Because these two cities are relatively near to each other, this may indicate a regional phenomenon.

Actual vs Apparent Daily Mean Temperatures by Month, All Cities, by Year

This chart aggregates all cities and shows the development of real and apparent daily mean temperatures over each individual year.



Observations:

- All years show a general pattern of apparent temperatures increasing during the summer months and falling to the same level as, or even below, real temperatures during the winter.
- From 2000-2009, 2011-2014 and from 2017-2019, apparent temperatures during the first two months of the year were lower than real temperatures.

General Seasonal Patterns by City

For the following charts, daily weather data for each individual city was aggregated by month and visualized as a time-series.

All charts for each city can be viewed in the folder 'visualizations/eda_individual_cities'.

Average Daily Minimum and Maximum Temperatures by Month

Observations:

- Delhi, Jaipur and Lucknow have the greatest range of daily temperatures out of all cities and the widest seasonal variation (coldest winters and hottest summers). These 3 cities also show the least difference between minimum and maximum temperatures.
- Mumbai and Ahmedabad's daily maximum temperatures experience more variation each year compared to those cities' minimum temperatures and compared to maximum temperatures in other cities.
- Bangalore shows the greatest difference between minimum and maximum temperatures, as well as the least amount of seasonal variation in both those values.

Average Daily Mean vs Apparent Daily Mean Temperatures by Month

Observations:

- Chennai, Mumbai and Kolkata show apparent temperatures being consistently higher than actual temperatures.
- Delhi, Jaipur and Lucknow show apparent temperatures that are mostly close to and, during the winter months, often below actual temperatures.
- All cities show higher apparent temperatures during the summer months. The exception to this, however, is Hyderabad in the years 2003 and 2008, where daily mean temperatures were higher than apparent ones even during summer.

Average Daily Precipitation Levels by Month

Observations

- Different cities experienced higher than usual peaks in precipitation levels at different points during this time period. E.g. Delhi in summer 2016, Jaipur in summer 2013, Chennai in autumn 2021 and Ahmedabad in summer 2017.

- There appears to be no difference between daily precipitation and daily rain levels, indicating that other forms of precipitation, such as snow or sleet, did not occur at all in any of these cities.

Average Daily Maximum Wind and Wind Gust Speeds by Month

Observations:

- Chennai, Mumbai, Pune and Kolkata show increased maximum wind speeds and wind-gust speeds from 2017 onwards, while the remaining cities did not show any similar change in wind speeds. This increase in wind intensity may be linked to the coastal or near-coastal location of these 4 cities.

Differences between Real and Perceived Daily Temperatures

This section analyses the differences between actual and apparent daily temperatures (mean, minimum and maximum). This value was created by subtracting actual temperatures from apparent temperatures for each day.

For each city, the following visualizations were created:

- Development of the differences between actual and apparent minimum and maximum daily temperatures (aggregated monthly) between 2000 and 2024.
- Development of the differences between actual and apparent daily mean temperatures (aggregated monthly) between 2000 and 2024.
- Regression plots showing the trend in these temperature differences over the years.

All charts for each city can be viewed in the folder 'visualizations/eda_individual_cities'.

Observations:

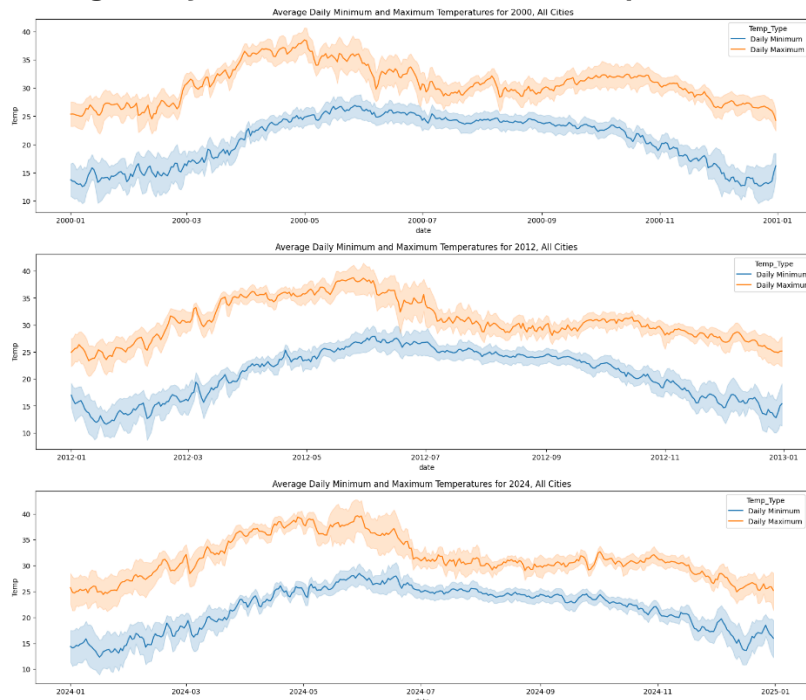
- Cities located in the South of the country (Chennai, Bangalore, Hyderabad) exhibit more fluctuation throughout each year. Mumbai and Pune exhibit a similar pattern. The remaining cities show a steadier rise and fall in temperature differences. This may be due to the difference in humidity and precipitation levels and their effect on perceived temperature.
- Apparent temperatures are rarely 2°C colder than actual temperatures. The city of Chennai shows apparent temperatures consistently higher than actual ones. Mumbai, Kolkata, Bangalore and Hyderabad show apparent temperatures that are occasionally 1°C colder than actual ones.
- Delhi, Lucknow and Jaipur show seasonal patterns of colder apparent temperatures at the beginning of every year.
- The difference between real and apparent daily mean temperatures in all cities appears to be increasing over the years, except for Pune, which shows a steady, or even slight decrease in temperature differences. The city of Chennai shows the largest increase in temperature difference over the years.

Seasonal Patterns for Selected Years, All Cities

The following visualizations combined daily records from all cities for the years 2000, 2012 and 2024.

The charts showing wind speeds also included wind direction as a variable.

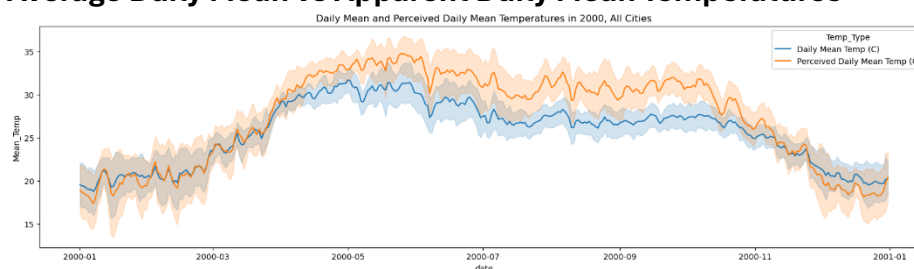
Average Daily Minimum and Maximum Temperature

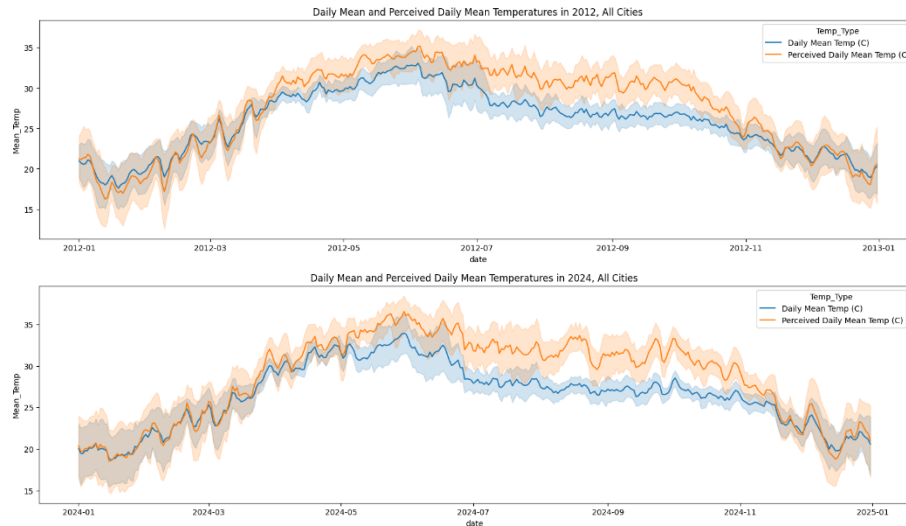


Observations:

- All 3 years show a narrowing of the difference between minimum and maximum temperatures during the summer months.
- Temperatures begin to rise earlier in the year in 2012 and 2024.
- The range of temperatures does not vary significantly among these years.

Average Daily Mean vs Apparent Daily Mean Temperatures

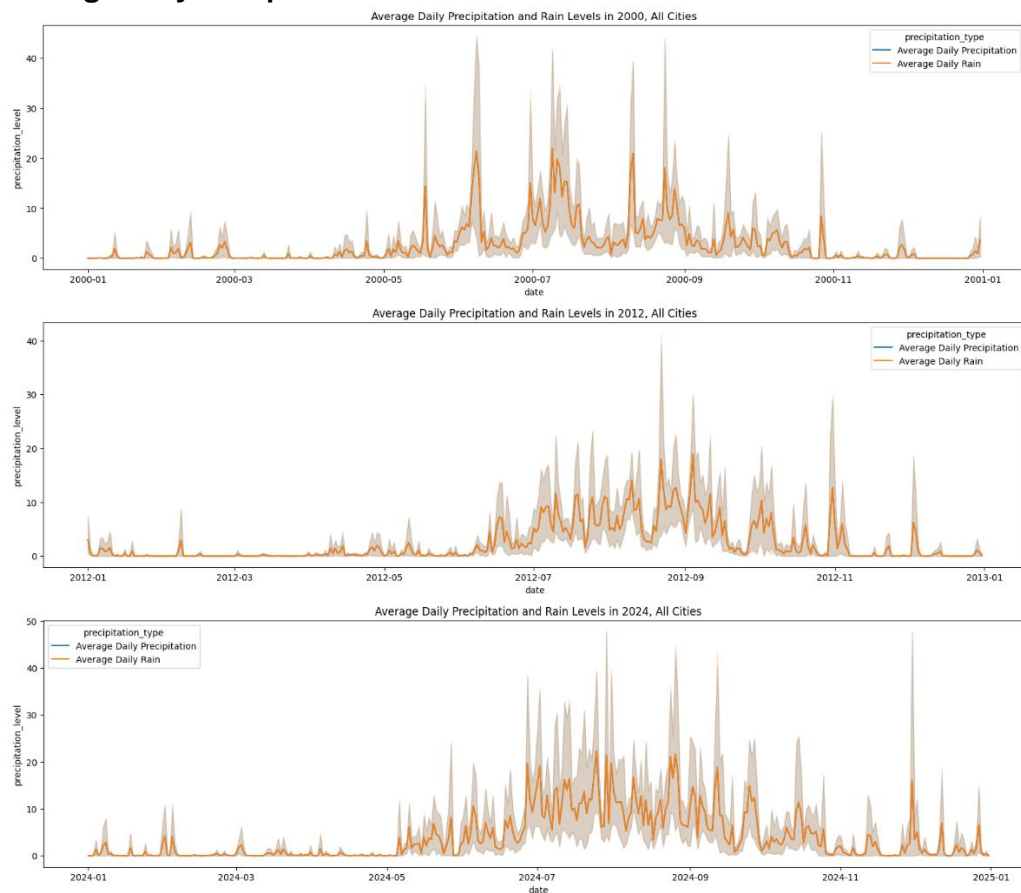




Observations:

- The number of days with perceived temperatures lower than real temperatures appears to decrease over the years.
- Perceived temperatures are most likely to be equal to or lower than actual temperatures during the winter months, between November and March.
- Changes in both temperature levels are mostly similar and simultaneous.

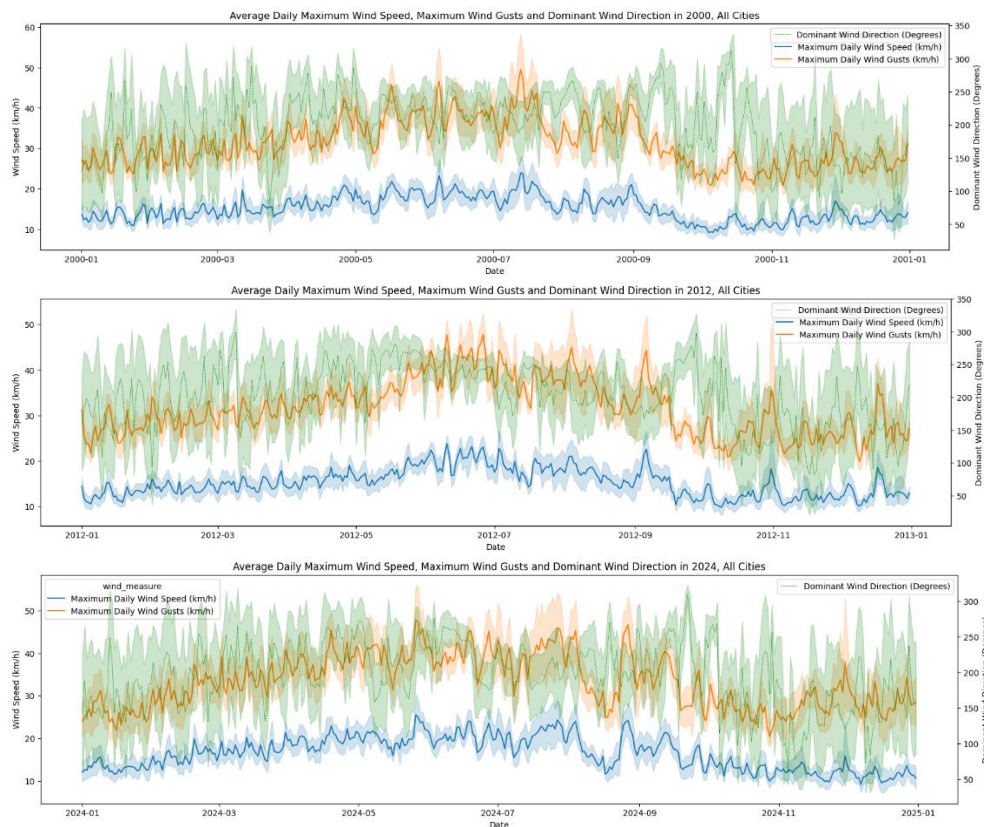
Average Daily Precipitation Levels



Observations:

- Precipitation levels peak between July and September and again in November. This second peak in precipitation is likely due to the later monsoon affecting some cities.
- May 2012 appears to have been drier than the same month in 2000 and 2024.
- Precipitation levels in 2024 were overall higher, with some daily levels nearing 50mm.

Average Daily Maximum Wind and Wind Gust Speeds; Dominant Wind Direction



Observations:

- Wind speeds and wind gust speeds peak during the summer months and again around November. This corresponds to increased rain levels.
- The highest winds in 2012 and 2024 had a direction of approximately 250°, while in 2000, the highest winds came from 200°.
- Wind direction may be more useful when looking at data from individual cities than from the entire group.

Differences between Real and Perceived Daily Temperatures

For the years 2000, 2012 and 2024, the following visualizations were created:

- Development of differences between actual and apparent daily minimum and maximum temperatures (all cities combined).
- Development of differences between actual and apparent daily mean temperatures (all cities combined).
- Regression plots showing the trend in these temperature differences (Degrees) over the years.

All charts for these selected years can be viewed in the folder 'visualizations/eda_individual_years'.

Observations:

- Years 2000 and 2012 showed more daily minimum, maximum and mean apparent temperatures cooler than actual temperatures.
- All daily temperatures appear warmer during the summer and fall months of the year (June to November), likely corresponding with increased precipitation levels from the monsoon.
- Actual and Perceived temperatures are almost the same during the winter months. Perceived temperatures cooler than actual ones are common during the winter and early spring months (January to April).
-

Relationships between Variables

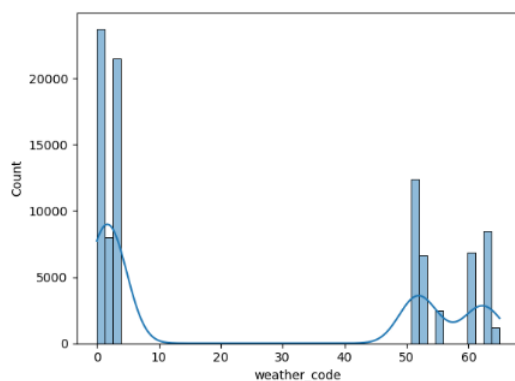
WMO Weather Codes

Each daily weather record contains a WMO weather code, a numeric value indicating certain weather conditions.

The numbers correspond to the following conditions:

{0: 'Cloud development not observed or not observable',
3: 'Clouds generally forming or developing',
63: 'Rain, not freezing, continuous, moderate at time of observation',
53: 'Drizzle, not freezing, continuous, moderate at time of observation',
2: 'State of sky on the whole unchanged',
1: 'Clouds generally dissolving or becoming less developed',
51: 'Drizzle, not freezing, continuous, slight at time of observation',
61: 'Rain, not freezing, continuous, slight at time of observation',
55: 'Drizzle, not freezing, continuous, heavy (dense) at time of observation',
65: 'Rain, not freezing, continuous, heavy at time of observation'}

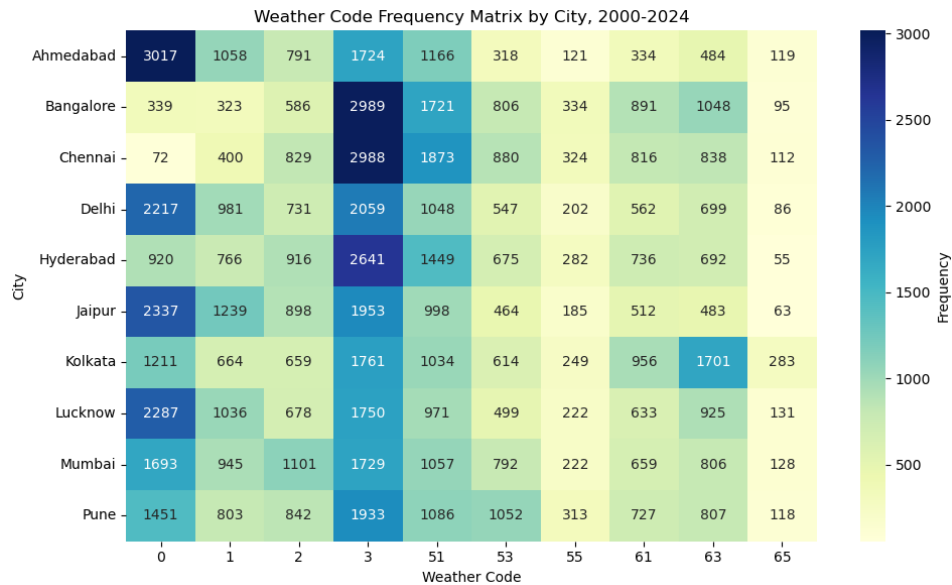
Frequency of Weather Codes in the Dataset



At least half of all records in this dataset show weather codes from 0 to 10, in this case, codes 0, 1, 2 and 3. This means that at least half the days recorded did not have any rain or precipitation.

Weather Code Frequency Matrix by City, 2000-2024

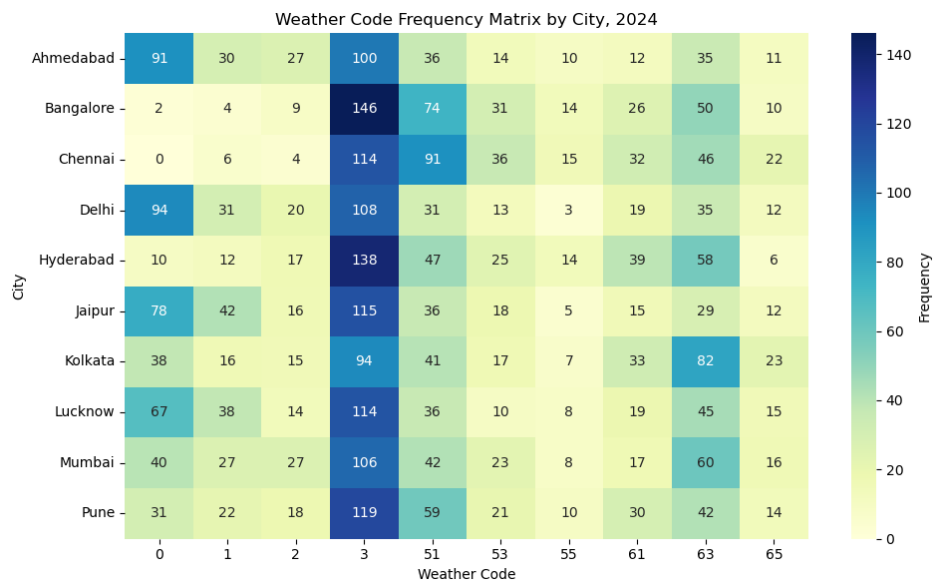
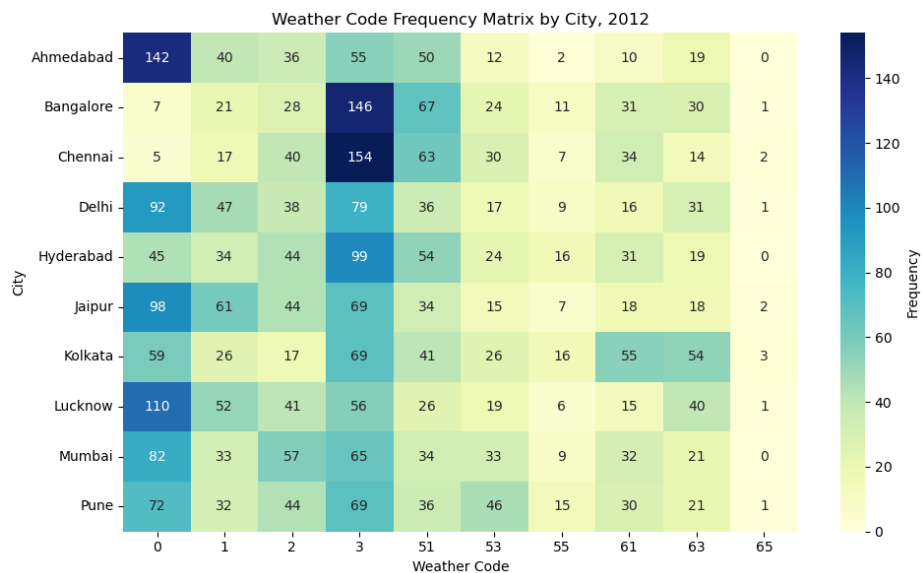
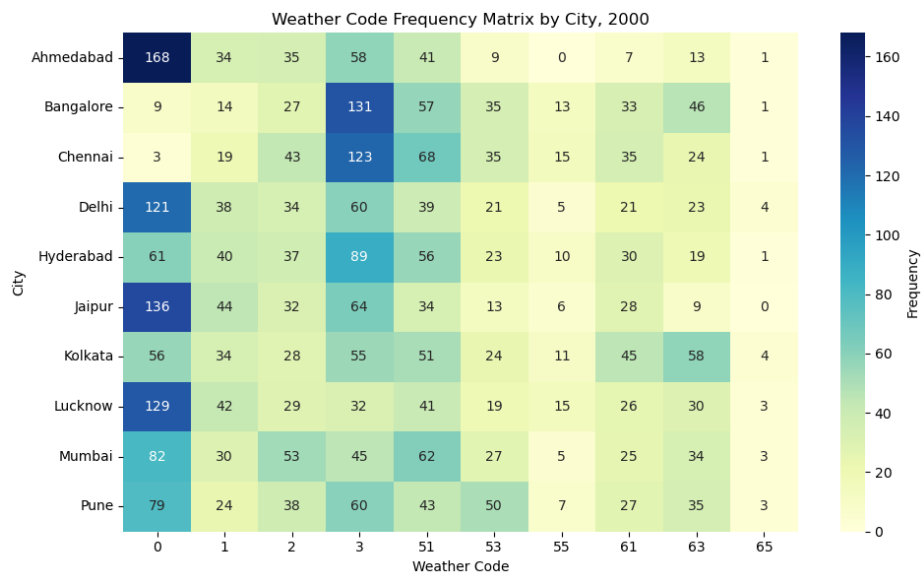
This shows the number of entries for each city containing a certain weather code.



Observations:

- Ahmedabad had the highest frequency of clear days (weather code 0), Chennai had the lowest frequency. This was also the most observed weather code for Delhi, Jaipur and Lucknow.
- Bangalore, Chennai and Hyderabad had the highest frequency of generally cloudy days (weather code 3). This was also the most observed weather code for Mumbai, Kolkata and Pune.
- Chennai, Bangalore and Hyderabad had the highest frequency of days with light drizzle (weather code 51).
- Weather codes 55 (heavy, continuous drizzle) and 65 (heavy, continuous rain) were the least common across all cities. Bangalore, Chennai and Pune showed the highest frequency of weather code 55, while Kolkata showed the highest frequency of weather code 65.
- Kolkata also showed the highest frequency of weather code 63 (moderate rain) compared to other cities.

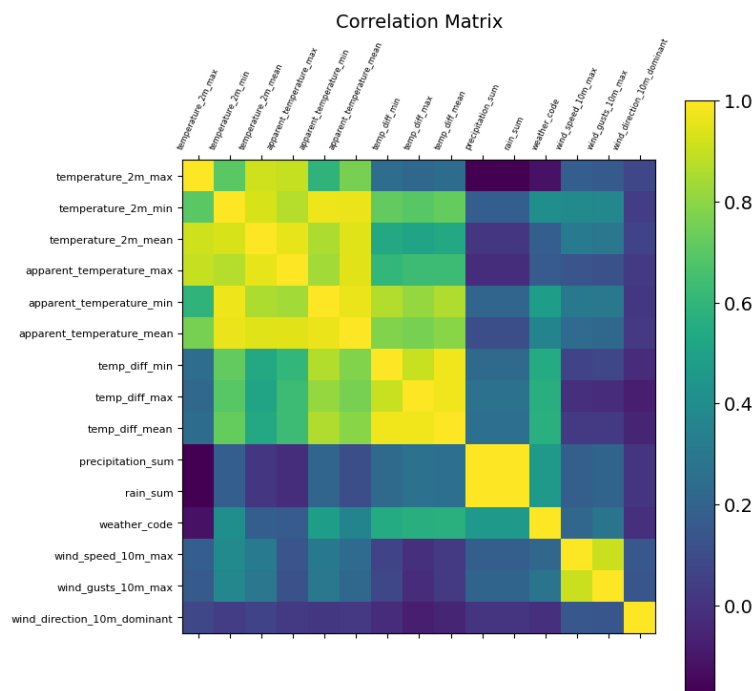
Weather Code Frequency Matrix, Selected Years



Observations:

- Ahmedabad's precipitation levels appear to have increased between 2000 and 2024, with fewer occurrences of weather code 0 (clear skies) and increasing numbers of days with cloudy and rainy conditions.
- Weather code 65 (heavy, continuous, non-freezing rain) appears to have increased in overall frequency between 2012 and 2024, with all cities showing higher counts of this weather code in 2024.

Correlation Matrix



The numbers in this correlation matrix represent the strength of the relationship between each variable. A correlation coefficient of 1 indicates a perfectly matching relationship, a coefficient of 0 indicates no relationship at all. A negative value indicates an inverse relationship.

Weather codes with higher numbers indicate, in this data, increasing clouds and precipitation.

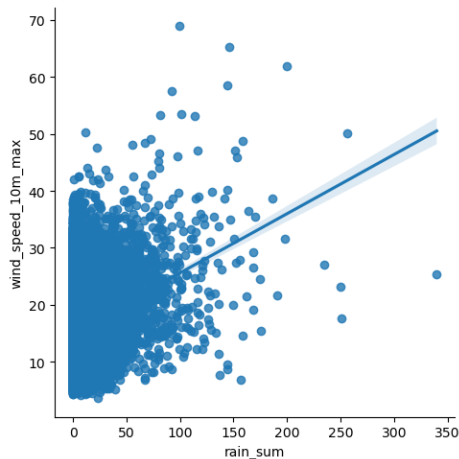
Observations

- Wind direction does not have any significant effect on other weather conditions nor on the weather code assigned.
- Maximum temperature does not appear to have any correlation with the weather code (possibly due to all the cities listed having a subtropical/tropical monsoon climate)
- The difference between apparent and actual temperature (the temp_diff variables) correlates relatively strongly with the weather code, indicating that increased precipitation increases apparent temperature.
- Precipitation and rain have the strongest correlation with weather code (again, due to higher numbers representing rainier conditions)
- Precipitation and rain correlate perfectly (again, due to the warm climate of the cities in this dataset, there is not likely to be other forms of precipitation such as snow, sleet, etc.)

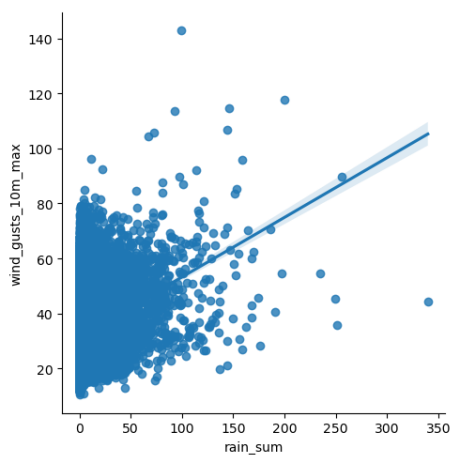
Relationships between Selected Variables

The following linear regression charts show relationships between selected variables across the dataset. The linear regression line shows the equation most strongly representing this relationship. The more dots fall on this line, the stronger the linear relationship.

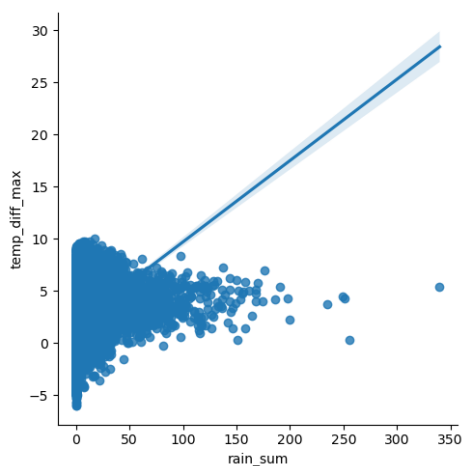
Rain (mm) and Maximum Wind Speed (km/h)



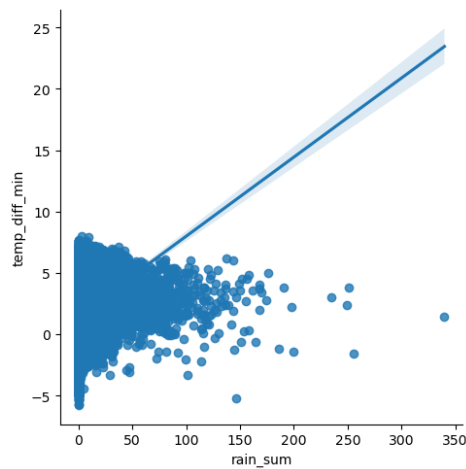
Rain (mm) and Max. Wind Gust Speed (km/h)



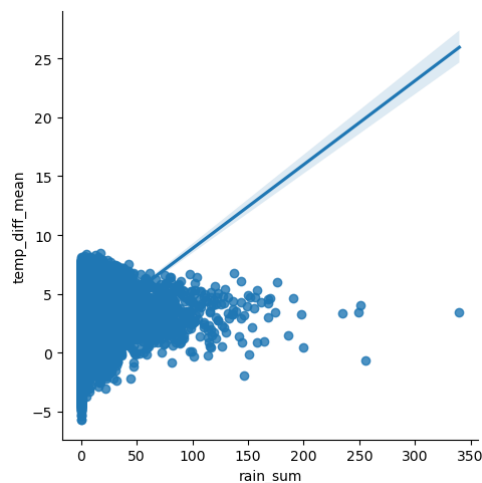
Difference between Apparent and Real Daily Maximum Temperatures (°C) and Rain (mm)



Difference between Apparent and Real Daily Minimum Temperatures (°C) and Rain (mm)



Difference between Apparent and Real Daily Mean Temperatures (°C) and Rain (mm)

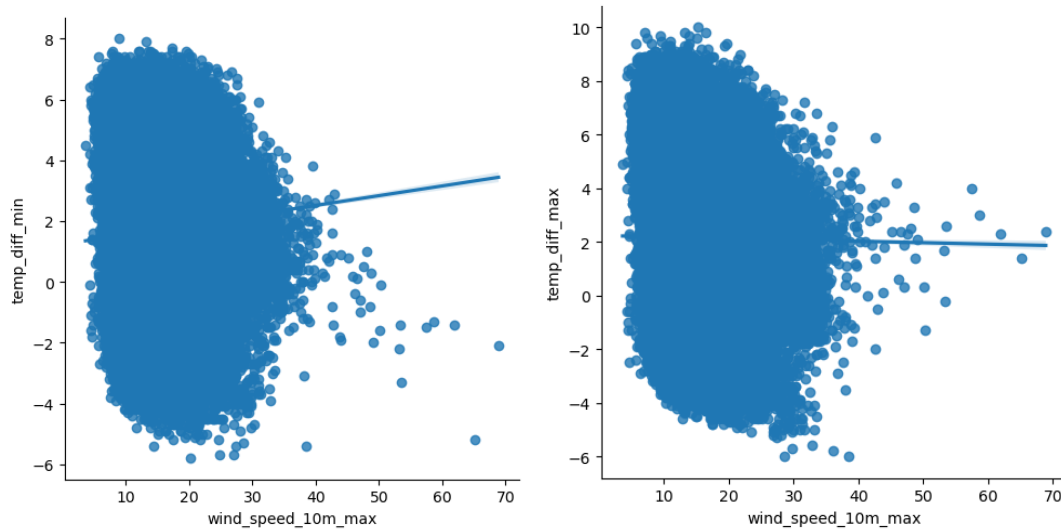


Observations

- All charts represent a weak linear relationship between variables. This means that linear regression cannot be used to generate accurate predictions for future values.
- All charts show a weak, positive correlation between variable pairs.
- The correlation between rain and other variables is particularly weak.
- Rain data is heavily skewed to the right, with the majority of records showing 0 rain.
- Further data on humidity levels and cloud cover would also be helpful in determining the cause of higher and lower apparent temperatures.

Relationship between Maximum Wind Speed (km/h) and Perceived Temperature Difference (°C)

The following charts attempt to establish a connection between daily maximum wind speed and the difference between apparent and real daily temperature levels.



Observations

- All correlations between wind speed and its effect on apparent temperature are very weak, as indicated by the majority of points falling away from the regression line.
- There is some positive correlation between wind speed and the difference between actual and apparent minimum temperatures. This indicates that higher wind speeds cause minimum temperatures to feel colder than they are.
- Increased wind speeds show a slight cooling effect on perceived temperatures, as indicated by the downward-sloping regression line. However, as with previously-shown regression charts, this relationship is not particularly strong.
- Linear regression charts may yield more significant results using data from individual cities than from the entire dataset. Using data for selected years may also help account for recurring seasonal patterns.

Key Insights

- Perceived daily temperatures appear to have increased from 2000 to 2024, with the city of Chennai showing the greatest increase in temperature.
- Cities with arid climates, such as Jaipur and Lucknow, show the lowest and highest temperatures, as well as the least difference between real and perceived temperature.
- Coastal cities that receive higher amounts of rain, such as Mumbai and Chennai, show less variation between minimum and maximum temperatures, as well as between winter and summer temperatures. Apparent temperatures in these cities are almost always higher than actual temperatures.
- Linear regression shows a weak relationship between precipitation levels themselves and real and perceived temperatures. This may be due to the fact that precipitation is highly seasonal across all cities, with distinct wet and dry seasons. Differences in temperatures and temperature differences between cities with different average levels of precipitation may be due to other, related factors, such as humidity or atmospheric pressure.
- Northern and Northwestern India experienced a severe heatwave between May 23 and May 30, 2024. The highest real and perceived temperatures were recorded in Ahmedabad and, a week later, in Delhi and Lucknow.
- The highest values for daily temperature, precipitation and wind speed all occurred within the last 7 years of the dataset. This suggests an increase in extreme weather events across India in recent years.
- The cities of Pune, Bangalore and Hyderabad did not experience any extreme weather during this period. This may be due to their geographical location far from the coast, or even due to features in their urban design that mitigate the effects of severe weather.

Next Steps

- Select and analyse individual years or cities where unusual or extreme weather events took place between 2000 and 2024. Determine if there is any connection between these events, as well as the likelihood of similar events taking place more frequently in the future.
- Predict future weather conditions using time-series forecasting. The exact model to use with this data depends on its stationarity (characteristics such as minimum, maximum and mean remain constant throughout the time-series) and its seasonality (the time-series shows recurring patterns at regular intervals). Because the data is both seasonal and relatively predictable, a SARIMA (seasonal autoregressive integrated moving average) model may likely be the most appropriate one to use with this data.
- The dataset contains WMO weather codes for each daily entry, indicating a set of conditions. We can use different machine-learning models, both supervised (e.g. k-Nearest Neighbours) or unsupervised (e.g. Random Forest Classifier), to classify daily weather records by weather code.

